

KYLE WILLIAMSON

+1 (647) 471-9495 | mgkshome7@gmail.com | Richmond Hill, ON, Canada
kawilliam.github.io/portfolio_website | linkedin.com/in/kyle-abraham-williamson | github.com/kawilliam

SUMMARY

Software engineering student building production systems from mobile apps to microservices, with active interest in low-level and systems programming. Seeking a backend, full-stack, or systems-oriented new-graduate role.

EDUCATION

York University - Lassonde School of Engineering

Sept 2021 - April 2027

Bachelor of Engineering, Software Engineering **Relevant Coursework:** Mission-Critical Systems, Software Design & Testing, Data Structures & Algorithms, Operating Systems, Computer Organization, Embedded Systems, Capstone Engineering Project

PROJECTS

BlockBid Auction System - Full-Stack Microservices Platform

Oct 2025 - Dec 2025

Docker / Spring Boot / WebSocket / Ethereum

- Architected 6-service microservices auction platform with database-per-service pattern, real-time WebSocket bidding with JWT/RBAC authentication, Ethereum blockchain audit trails, and 30+ REST endpoints, deployed via Docker Compose with a single command.

Runz - Street Basketball Ranking Mobile App

Jan 2026 - Present

Flutter / Supabase / Riverpod / PostGIS / Deno

- Architected full-stack competitive 1v1 basketball platform targeting iOS and Android with real-time game sessions via Supabase Realtime WebSocket subscriptions, eliminating polling for live status updates across 12+ Postgres tables during active matches.
- Designed dual-implementation Elo ranking engine (Dart + PL/pgSQL) with dynamic K-factors, 5-tier progressive ranking system, GPS verification weighting, and anti-cheat protections including streak cooldowns, rep quarantine, and server-side proximity validation via PostGIS.
- Built 3 Edge Functions (TypeScript/Deno) supporting 15+ push notification types via OAuth2-signed FCM v1 API, persistent offline action queue with exponential-backoff retry, and cron-driven auto-resolution for abandoned game sessions, backed by unit and widget tests.

YorkU Conference Room Scheduler

Sept 2025 - Dec 2025

Java / Design Patterns / UML

- Led team of 4 in building enterprise room booking system with conflict-free scheduling, role-based access control, automated waitlisting, and comprehensive UML documentation, applying Observer, Factory, and Strategy design patterns.

GTA Real Estate Hotspots - Spatial Network Analysis

Sept 2025 - Dec 2025

Python / PyTorch / NetworkX

- Analyzed 358,713 building permits from Toronto Open Data using graph-based spatial network of 98 regions and 165 connections, with Spatial Autoregressive model achieving 44.7% lower RMSE ($p = 0.037$) and 60% precision in identifying top-10 real estate growth hotspots.

Terminal Portfolio - Interactive Developer Website

April 2025

Vanilla JS / Web Audio API / CSS3

- Built interactive terminal emulator with custom shell (14+ commands, arrow-key history, inactivity-triggered auto-demo, recruiter skip-path via URL parameter), procedural Web Audio API sound engine synthesizing POST beeps and keystroke clicks from oscillator nodes with zero audio files, and CRT-era aesthetic via layered CSS gradients and keyframe animations, all vanilla JS, HTML5, CSS3, zero dependencies.

PROFESSIONAL EXPERIENCE

Aritzia - Risk Associate

Nov 2025 - Feb 2026

Vaughan, ON

- Conducted inventory audits and fraud monitoring across store operations, coaching team members on loss prevention procedures and inventory accuracy.

Northern Shield Security - Security Professional

Nov 2024 - Present

Richmond Hill, ON

- Performed threat assessments and maintained incident documentation for high-value retail environments, coordinating security response with mall security teams.

Apple - Sales Specialist

Oct 2023 - April 2024

Toronto, ON

- Translated complex technical concepts for 30-50 customers daily across Apple's product ecosystem while completing ongoing technical certification training.

SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, Dart, SQL, SystemVerilog, HTML/CSS

Frameworks: Spring Boot, React.js, Flutter, Riverpod, Node.js, Express.js, Streamlit

Data & ML: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, NetworkX, Jupyter Notebooks

Infrastructure: Git, Docker, Docker Compose, Maven, Unix/Linux/macOS, CI/CD (GitHub Actions)

Databases: SQLite, MySQL, PostgreSQL, Supabase

APIs & Protocols: REST APIs, WebSocket, SSE, JWT, OAuth, Discord/Telegram Bot APIs

Tools: Postman, Agile, Jira, JUnit, Playwright, Selenium, Unix Shell, Data Structures & Algorithms, Excel/Numbers/Sheets

Interests: Embedded Systems, Low-Level Systems, FPGA, Firmware, VHDL, x86, Deep Learning, AI/GPU, Azure, MongoDB, Django, Windows PowerShell, Perl, TCL, Ethernet, CAD, Mathematics, Specification, Gaming

Spoken Languages: French, Italian